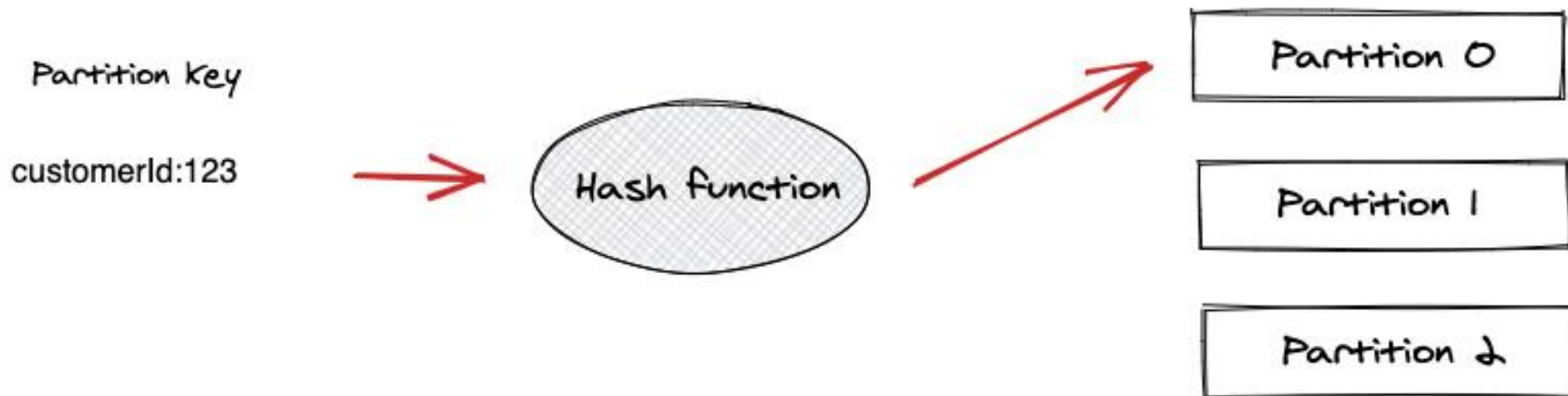


Kafka producers

- ❖ **Producers** write to and **Consumers** read from **Topic(s)**
- ❖ **Topic** associated with a log which is data structure on disk
- ❖ **Producer(s)** append **Records** at end of Topic log
- ❖ Topic **Log** consist of Partitions
 - ❖ Spread to multiple files on multiple nodes
- ❖ **Consumers** read from Kafka at their own cadence
 - ❖ Each Consumer (Consumer Group) tracks offset from where they left off reading
- ❖ **Partitions** can be distributed on different machines in a cluster
 - ❖ high performance with horizontal scalability and failover with replication

Kafka Producers

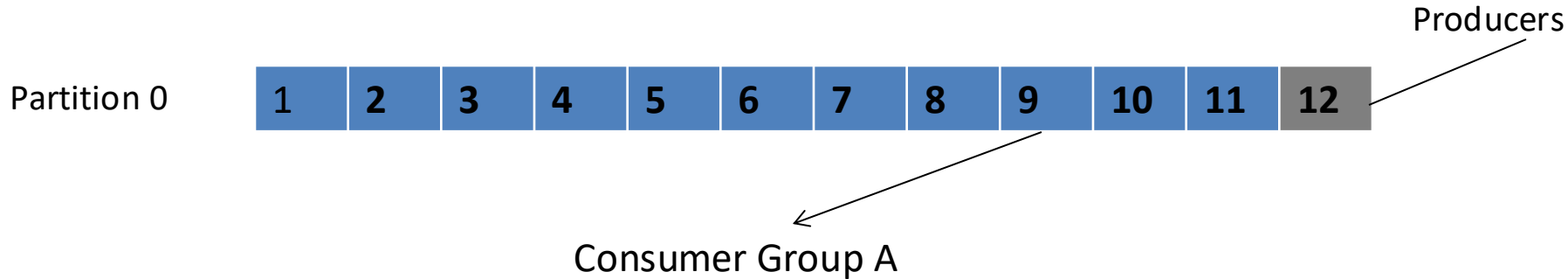
Tos



- ❖ **Producers** send records to topics
- ❖ **Producer** picks which partition to send record to per topic
 - ❖ Can be done in a **round-robin**
 - ❖ Can be based on priority
 - ❖ Typically based on **key** of **record**
 - ❖ Kafka **default partitioner** for Java uses hash of keys to choose partitions, or a round-robin strategy if no key
- ❖ Important: *Producer picks partition*

Kafka producers and consumers

Tos



Producers are writing at Offset 12
Consumer Group A is Reading from Offset 9.

- ❖ **Producers** write at their own cadence so order of Records cannot be guaranteed across partitions
- ❖ **Producer** configures consistency level (ack=0, ack=all, ack=1)
- ❖ **Producers** pick the **partition** such that Record/messages goes to a given same partition based on the data
 - ❖ Example have all the events of a certain 'employee Id' go to same partition
 - ❖ If order within a partition is not needed, a 'Round Robin' partition strategy can be used so Records are evenly distributed across partitions.

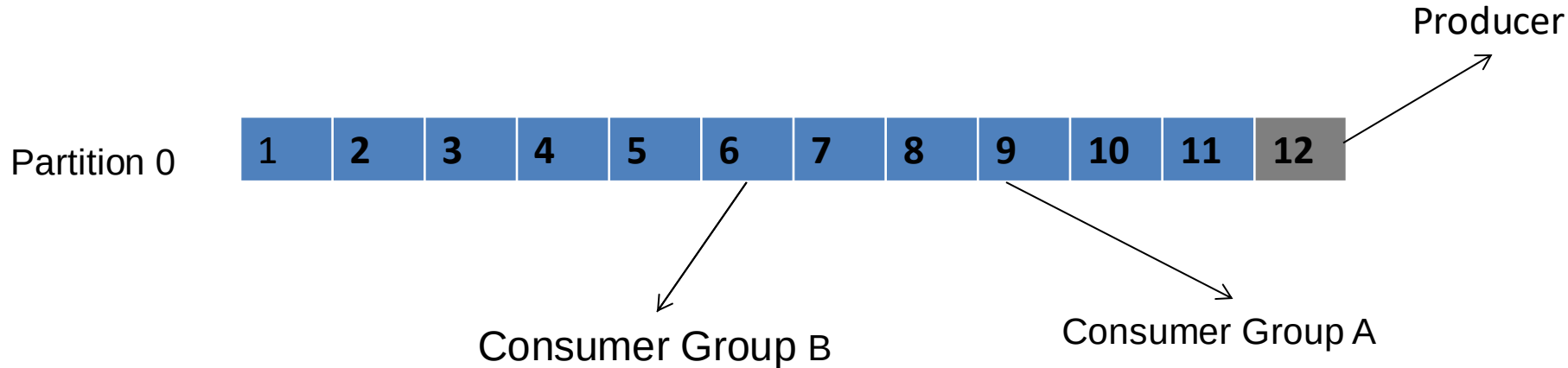
Kafka Consumers

- ❖ Consumers are grouped into a **Consumer Group**
- ❖ **Consumer group** has a unique id
- ❖ Each **consumer group** is a subscriber
- ❖ Each **consumer group** maintains its own offset
- ❖ Multiple subscribers = multiple consumer groups
- ❖ Each has different function: one might delivering records to micro services while another is streaming records to Hadoop
- ❖ **A Record** is delivered to one **Consumer** in a **Consumer Group**
- ❖ Each consumer in consumer groups takes records and only one consumer in group gets same record
- ❖ Consumers in Consumer Group **load balance** record consumption

- ❖ Kafka **Consumer** consumption **divides** partitions over consumers in a Consumer Group
- ❖ Each **Consumer** is exclusive consumer of a "**fair share**" of partitions
- ❖ This is Load Balancing
- ❖ **Consumer** membership in **Consumer Group** is handled by the Kafka protocol dynamically
- ❖ If new Consumers **join** Consumer group, it gets a share of partitions
- ❖ If Consumer **dies**, its partitions are split among remaining live Consumers in Consumer Group

Kafka Consumer Groups

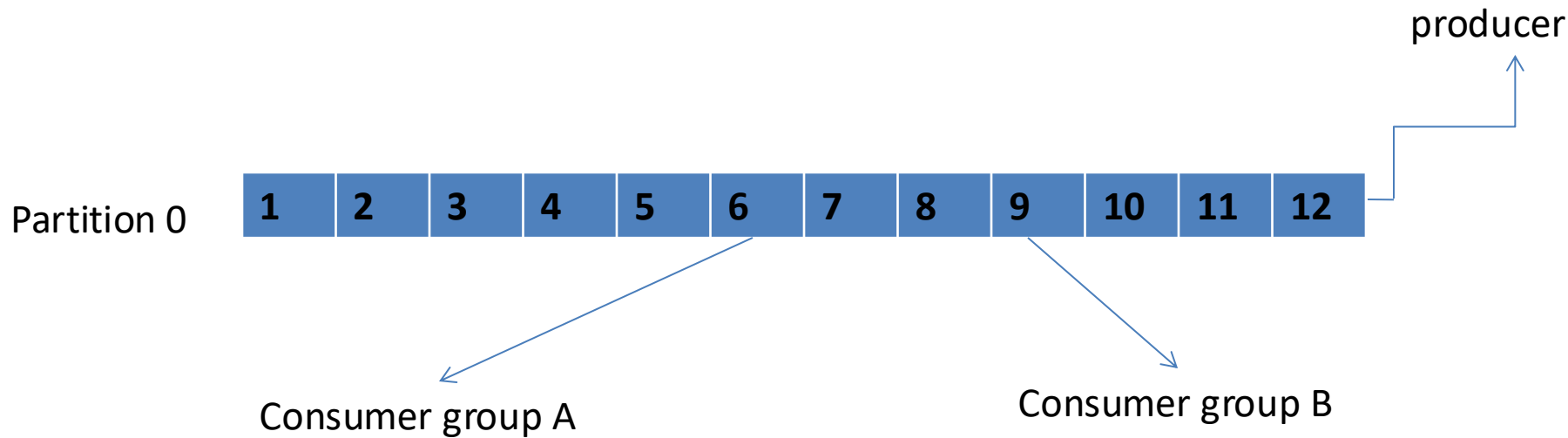
Tos



Consumers remember offset where they left off.
Consumers groups each have their own offset per partition

Kafka Topic Partition, Consumers, Producers

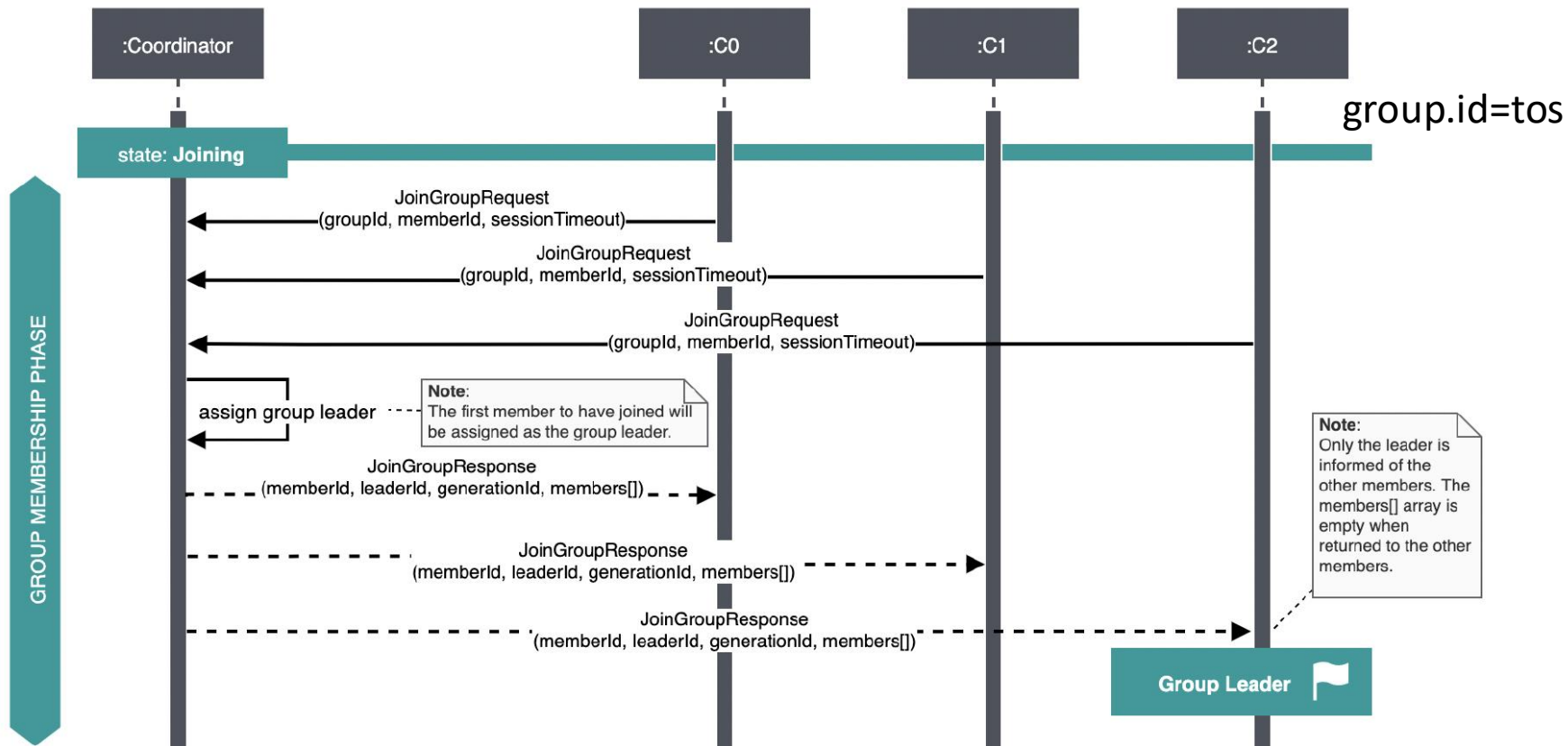
Tos



Consumer groups remember offset where they left off. Consumers groups each have their own offset. Producer writing to offset 12 of partition0 while... consumer Group A is reading from offset 6. Consumer Group B is reading from offset 9.

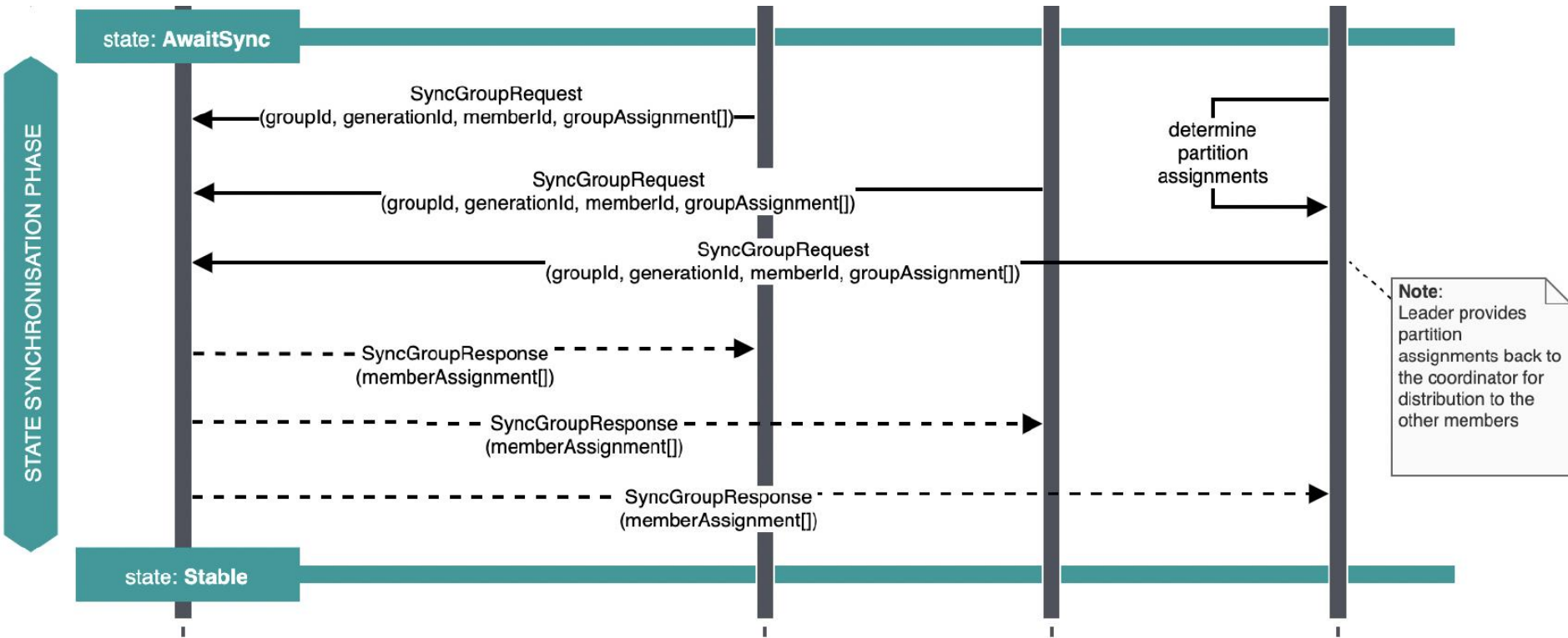
Kafka Consumer Groups Processing

Tos



Kafka Consumer Groups Processing

Tos



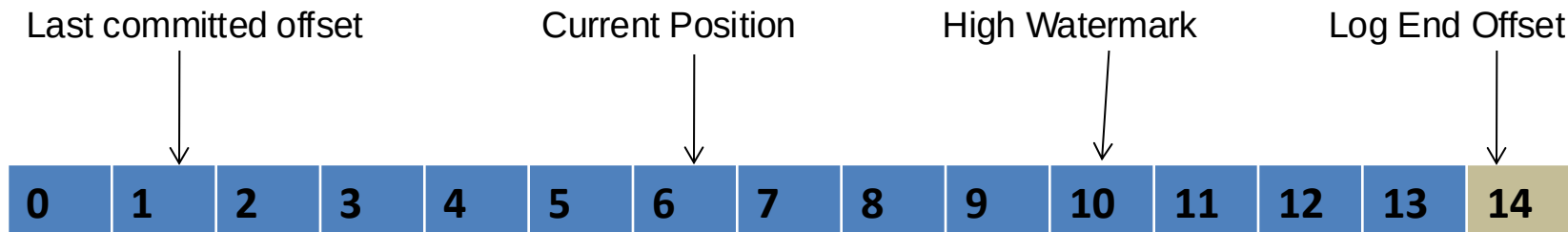
- ❖ How does Kafka divide up topic so multiple **Consumers** in a **Consumer Group** can process a topic?
- ❖ You group consumers into consumers group with a group id
- ❖ **Consumers** with same id belong in same **Consumer Group**
- ❖ One **Kafka broker** becomes **group coordinator** for Consumer Group
 - ❖ assigns partitions when new members arrive (older clients would talk direct to ZooKeeper now broker does coordination)
 - ❖ or reassign partitions when group members leave or topic changes (config / meta-data change)
- ❖ When **Consumer group** is created, offset set according to reset policy of topic

- ❖ **Consumers** notify broker when it successfully processed a record
 - ❖ advances offset
- ❖ If **Consumer** fails before sending commit offset to Kafka broker,
 - ❖ different **Consumer** can continue from the last committed offset
 - ❖ some Kafka records could be reprocessed
 - ❖ *at least once behaviour*
 - ❖ *messages should be idempotent*

- ❖ Kafka stores offsets in topic called “__consumer_offsets”
- ❖ Uses Topic Log Compaction
- ❖ When a consumer has processed data, it should commit offsets
- ❖ If consumer process dies, it will be able to start up and start reading where it left off based on offset stored in “__consumer_offset”

Kafka Consumer: What can be consumed?

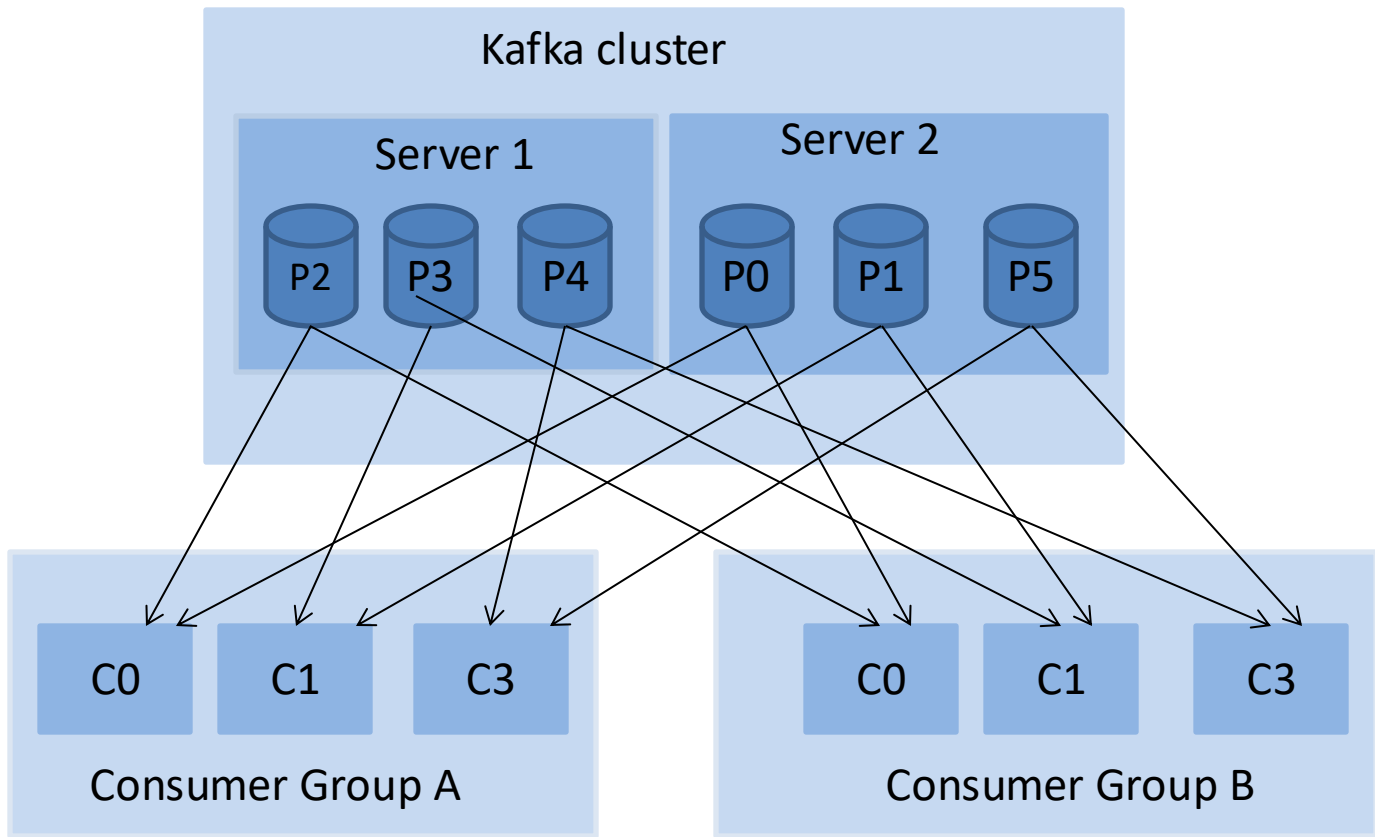
- ❖ **"Log end offset"** is offset of last record written to log partition and where **Producers** write to next
- ❖ **"High watermark"** is offset of last record successfully replicated to all partitions followers
- ❖ **Consumer** only reads up to "high watermark".
Consumer can't read un-replicated data



- ❖ Only a single **Consumer** from the same **Consumer Group** can access a single **Partition**
- ❖ If **Consumer in a Group** count **exceeds** Partition count:
 - ❖ Extra Consumers remain idle; can be used for failover
- ❖ If more Partitions than Consumer in a Group instances,
 - ❖ Some Consumers will read from more than one partition

2 server Kafka cluster hosting 6 partitions(P0-P5)

Tos



- ❖ You can run more than one Consumer in a JVM process
- ❖ If processing records takes a while, a single Consumer can run multiple threads to process records
 - ❖ Harder to manage offset for each Thread/Task
 - ❖ One Consumer runs multiple threads
 - ❖ 2 messages on same partitions being processed by two different threads
 - ❖ Hard to guarantee order without threads coordination
- ❖ **PREFER:** Multiple Consumers can run each processing record batches in their own thread
 - ❖ Easier to manage offset
 - ❖ Each Consumer runs in its thread
 - ❖ Easier to manage failover (each process runs X num of Consumer threads)

Run Kafka Producer

Tos

 start-producer-console.sh x

```
1  #!/usr/bin/env bash
2  cd ~/kafka-training
3
4  kafka/bin/kafka-console-producer.sh --broker-list \
5  localhost:9092 --topic my-topic
```

Run Kafka Consumer

Tos

 start-consumer-console.sh x

```
1  #!/usr/bin/env bash
2  cd ~/kafka-training
3
4  kafka/bin/kafka-console-consumer.sh --bootstrap-server localhost:9092 \
5  --topic my-topic --from-beginning
```

Running Kafka Producer and Consumer

Tos

```
new-employees
~/kafka-training/lab1/solution
[$ ./start-producer-console.sh
This is message 1
This is message 2
This is message 3
Message 4
Message 5
Message 6
Message 7
□
```

```
Last login: Sat May 13 13:57:09 on ttys004
~/kafka-training/lab1/solution
[$ ./start-consumer-console.sh
Message 4
This is message 2
This is message 1
This is message 3
Message 5
Message 6
Message 7
□
```

Lab : Basic Kafka Operations - CLI (Producer & Consumer) – 30 Mins